Assignment-8.R

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Problem on P-value Extravaganza.

a) Which of the "ASA Statement on Statistical Significance and P-Values" has not been carefully read by the guy on the hammock?

->he stated that the ‘ p-value tells us how likely the null hypothesis is true ’. so, he didn’t read the 2nd point of the principles of the ASA statement on the P-values which is P-values do not measure the probability that the studied hypothesis is true or the probability that the data were produced by random chance alone. **But It is a statement about data in relation to a specified hypothetical explanation, not about the explanation itself.**

b) At a certain point, Sir Fisher adopts the Neyman-Pearson approach. Tells at which minute of the video it approximately happens.

-> According to the video, Sir Fisher adopts the Neyman-Pearson approach at **7:45** minutes where he explains firstly to choose the **“α” %value (rejection region)** and then explains to choose **one/two-tail tests using the given data**.

c) At minute 9.33, a p-value is computed. Is that a p-value for a one-tail or a two-tail test?

-> At minute 9.33, the P-value is computed in a **one-tail test** because our null hypothesis with a given distribution of mean is zero. our p-value of the distribution is under the null hypothesis. So, one tail test is used here.

d) Find a mistake in the formula at minute 9.51.

->in minute 9.51 they gave p-value(0.2)= 0.4575>0.5. according to the equation, we have **the value 0.5 which is 50%** is less than 45% which is wrong to interpret the p-value. We usually use the “α” value to interpret the result of the p-value. Generally, the α value is 0.05 which is 5% where **0.05<0.4575** so there is no evidence against the null hypothesis.

```r
# 5.8 #
library(alr4)
```

```r
data(cakes)
m1 <-lm(Y~X1+X2+I(X1^2)+I(X2^2)+X1:X2, data=cakes)
summary(m1)

##
## Call:
## lm(formula = Y ~ X1 + X2 + I(X1^2) + I(X2^2) + X1:X2, data = cakes)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -0.4912 -0.3080  0.0200  0.2658  0.5454
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.204e+03  2.416e+02  -9.125 1.67e-05 ***
## X1           2.592e+01  4.659e+00   5.563 0.000533 ***
## X2           9.918e+00  1.167e+00   8.502 2.81e-05 ***
## I(X1^2)     -1.569e-01  3.945e-02  -3.977 0.004079 **
## I(X2^2)     -1.195e-02  1.578e-03  -7.574 6.46e-05 ***
## X1:X2       -4.163e-02  1.072e-02  -3.883 0.004654 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4288 on 8 degrees of freedom
## Multiple R-squared:  0.9487, Adjusted R-squared:  0.9167
## F-statistic:  29.6 on 5 and 8 DF,  p-value: 5.864e-05
```

According to the summary given absolute t-value is more than 2. so, our model is significant also the p-values are less than 0.05% so our model is definitely significant.

```
# B) here I need to add the regressor block to the given regression function
m2<-lm(Y~X1+X2+I(X1^2)+I(X2^2)+X1:X2+block, data=cakes)
summary(m2)
```

```
##
## Call:
## lm(formula = Y ~ X1 + X2 + I(X1^2) + I(X2^2) + X1:X2 + block,
##     data = cakes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4525 -0.3046  0.0200  0.2924  0.4883
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.205e+03  2.542e+02  -8.672 5.43e-05 ***
## X1           2.592e+01  4.903e+00   5.287 0.001140 **
## X2           9.918e+00  1.228e+00   8.080 8.56e-05 ***
## I(X1^2)     -1.569e-01  4.151e-02  -3.779 0.006898 **
## I(X2^2)     -1.195e-02  1.660e-03  -7.197 0.000178 ***
## block1       1.143e-01  2.412e-01   0.474 0.650014
## X1:X2       -4.163e-02  1.128e-02  -3.690 0.007754 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4512 on 7 degrees of freedom
```

```
## Multiple R-squared:  0.9503, Adjusted R-squared:  0.9077
## F-statistic: 22.31 on 6 and 7 DF,  p-value: 0.0003129
```

According to the new summary, the regressor block has an absolute t-value of 0.47 less than 2 which is not significant also p-value is 65% which shows our model is not significant. but in both cases, I cannot explain the significance surely because the sample size of the data is too low.

```
#problem 6.9
# 1)
h0 <-lm(Y~X1+X2+I(X1^2)+I(X2^2), data=cakes)
summary(h0)

##
## Call:
## lm(formula = Y ~ X1 + X2 + I(X1^2) + I(X2^2), data = cakes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.1565 -0.3615  0.0200  0.3285  1.1737
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.695e+03  3.247e+02  -5.219 0.000550 ***
## X1           1.135e+01  4.423e+00   2.566 0.030404 *
## X2           8.461e+00  1.769e+00   4.784 0.000996 ***
## I(X1^2)     -1.569e-01  6.317e-02  -2.483 0.034791 *
## I(X2^2)     -1.195e-02  2.527e-03  -4.730 0.001075 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6866 on 9 degrees of freedom
## Multiple R-squared:  0.852,  Adjusted R-squared:  0.7863
## F-statistic: 12.96 on 4 and 9 DF,  p-value: 0.0008913

# HA is
summary(m1)

##
## Call:
## lm(formula = Y ~ X1 + X2 + I(X1^2) + I(X2^2) + X1:X2, data = cakes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4912 -0.3080  0.0200  0.2658  0.5454
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -2.204e+03  2.416e+02  -9.125 1.67e-05 ***
## X1           2.592e+01  4.659e+00   5.563 0.000533 ***
## X2           9.918e+00  1.167e+00   8.502 2.81e-05 ***
## I(X1^2)     -1.569e-01  3.945e-02  -3.977 0.004079 **
## I(X2^2)     -1.195e-02  1.578e-03  -7.574 6.46e-05 ***
## X1:X2       -4.163e-02  1.072e-02  -3.883 0.004654 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4288 on 8 degrees of freedom
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## F-statistic:  29.6 on 5 and 8 DF,  p-value: 5.864e-05
```

```
qt(1-0.5,1,8)
```

11.81859

The quantile value is 11.8. f value is 4.8 which is less than the quantile so we reject the null hypothesis also for the m1(HA) summary we can study the p values of coefficients β5(X1:X2) also for β2 is significant because of the p-value which is nearly zero less than 0.05.so every coefficient is significant so there is no evidence to say β5 and β2 are zero so we use the interaction model.

```
#2)new H0
Hzero <-lm(Y~X2++I(X2^2), data=cakes)
summary(Hzero)
```

```
##
## Call:
## lm(formula = Y ~ X2 + +I(X2^2), data = cakes)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.30385 -0.00885  0.31129  0.62782  0.78681
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.431e+03  4.590e+02  -3.117   0.0098 **
## X2           8.124e+00  2.623e+00   3.097   0.0102 *
## I(X2^2)     -1.147e-02  3.747e-03  -3.060   0.0108 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.021 on 11 degrees of freedom
## Multiple R-squared:  0.5999, Adjusted R-squared:  0.5271
## F-statistic: 8.246 on 2 and 11 DF,  p-value: 0.006488
```

```
qt(1-0.5,3,8)
```

```
8.984022
```

The value of the quantile is 8.98 and the f value is 3.7 which is less than the quantile so we reject the null hypothesis according to the p-value the values are in no man's land but according to f test our test is significant mean we don't consider h0 as beta values zero.

problem (6.9)

① F-Test :- $$\frac{(RSS_{NH} - RSS_{AH}) / (df_{NH} - df_{AH})}{RSS_{AH} / df_{AH}}$$

$$\Rightarrow \frac{(0.6866 - 0.4288)/(9-8)}{0.4288/8}$$

$$\Rightarrow \frac{0.2578/1}{0.4288/8} = \frac{0.2578}{0.0536} = 4.8$$

~~F-statistic = 4.8~~

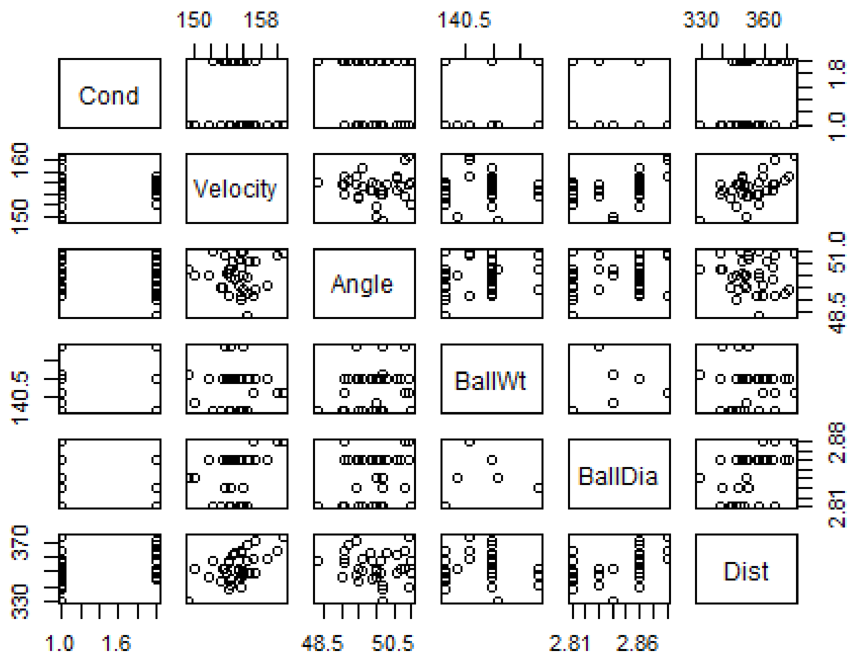
② F-Test :- $$\frac{\cancel{0.}\,(1.021 - 0.4288)/(11-8)}{0.4288/8}$$

$$= \frac{0.5922/3}{0.0536} = \frac{0.1974}{0.0536} = 3.7$$

```
#problem 5.20.1
head(domedata)

##   Cond Velocity Angle  BallWt BallDia  Dist
## 1 Head    149.3  50.2 141.100    2.84 329.3
## 2 Head    149.9  50.0 140.330    2.84 351.5
## 3 Head    153.2  49.5 140.099    2.81 343.9
## 4 Head    154.1  50.1 140.099    2.81 356.8
## 5 Head    155.9  50.0 140.099    2.81 350.1
## 6 Head    154.1  49.1 140.980    2.86 355.8

pairs(~Cond+Velocity+Angle+BallWt+BallDia+Dist,data=domedata)
```




```
m3=lm(Dist~Cond+Velocity+Angle+BallWt+BallDia,data=domedata)
summary(m2)

##
## Call:
## lm(formula = Y ~ X1 + X2 + I(X1^2) + I(X2^2) + X1:X2 + block,
##     data = cakes)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -0.4525 -0.3046  0.0200  0.2924  0.4883
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.205e+03  2.542e+02  -8.672 5.43e-05 ***
## X1           2.592e+01  4.903e+00   5.287 0.001140 **
## X2           9.918e+00  1.228e+00   8.080 8.56e-05 ***
## I(X1^2)     -1.569e-01  4.151e-02  -3.779 0.006898 **
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## block1       1.143e-01  2.412e-01   0.474 0.650014
## X1:X2       -4.163e-02  1.128e-02  -3.690 0.007754 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4512 on 7 degrees of freedom
## Multiple R-squared:  0.9503, Adjusted R-squared:  0.9077
## F-statistic: 22.31 on 6 and 7 DF,  p-value: 0.0003129
```

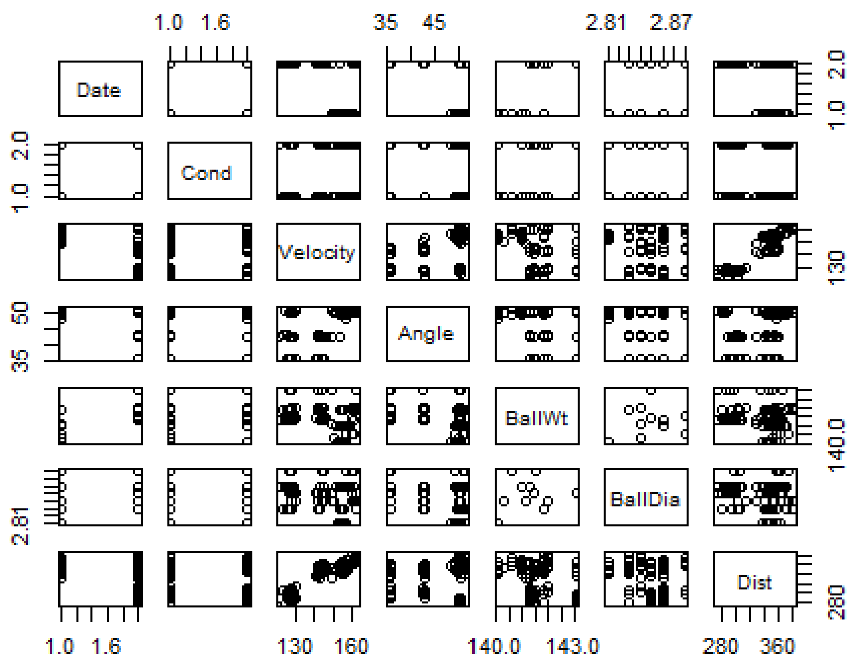
The fan does affect the ball distance because in summary, the absolute t-value is more than 2 for factors-condition, velocity, and ball diameter also the p-value is approximately less than α=0.05 value for the same regressors which means that these regressors are significant by the given model. so, The distance can be changed by the condition, velocity, and ball diameter.

5.20.2

```
head(domedata1)

##    Date     Cond Angle Velocity  BallWt BallDia  Dist
## 1 March Headwind  49.1    154.1 140.980    2.86 355.8
## 2 March Headwind  49.1    157.1 140.567    2.88 358.8
## 3 March Headwind  49.4    155.8 141.855    2.83 347.6
## 4 March Headwind  49.5    153.2 140.099    2.81 343.9
## 5 March Headwind  49.6    158.8 140.980    2.86 359.3
## 6 March Headwind  49.9    156.3 140.980    2.86 356.8

pairs(~Date+Cond+Velocity+Angle+BallWt+BallDia+Dist,data=domedata1)
```




```
m4=lm(Dist~Date+Cond+Velocity+Angle+BallWt+BallDia,data=domedata1)
summary(m4)

##
## Call:
## lm(formula = Dist ~ Date + Cond + Velocity + Angle + BallWt +
##     BallDia, data = domedata1)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.304  -6.332  -1.867   6.010  23.126
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  336.04115  267.61326   1.256   0.2125
## DateMay        8.61015    3.28563   2.621   0.0103 *
## CondTailwind   0.76762    1.90389   0.403   0.6878
## Velocity       2.63480    0.09289  28.364  < 2e-16 ***
## Angle         -1.36377    0.20458  -6.666 2.13e-09 ***
## BallWt        -2.85190    1.71414  -1.664   0.0997 .
## BallDia       27.29638   52.07840   0.524   0.6015
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.996 on 89 degrees of freedom
## Multiple R-squared:  0.9195, Adjusted R-squared:  0.914
## F-statistic: 169.4 on 6 and 89 DF,  p-value: < 2.2e-16
```

When we take into account the date in the summary for the factors of velocity and angle have significance as though our 2 experiments' results are different. The date does affect the result of distance which means the way to manipulate the fan